Internship Report - Supervised by Raphaël Thabut

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Introduction

This report presents the work carried out during a six-week internship at **CREST–ENSAE** under the supervision of **Raphaël Thabut (PhD)**. The project explores the idea that equity markets do not behave in the same way at all times, but instead follow different such as stability, crisis, or recovery. The main objective was therefore to study volatility and dependence across assets in a regime-aware framework. The results do not constitute a definitive proof that the chosen regime classification is optimal, but they provide several consistent indications that regimes matter and that this type of approach is useful.

An important part of the work focuses on a regime-aware GARCH model. By estimating volatility parameters separately for each regime and using them to standardize returns, I was able to build regime-specific correlation matrices across indices and sectors. This shows the expected rise and homogenization of correlations during crises, and also highlights how standardization clarifies the underlying structure beyond simple volatility effects. To reduce biases, I also fitted GARCH models locally within regimes, which confirmed the differences between periods.

The study also includes baseline ARCH(1) and GARCH(1,1) models fitted by regime, together with standard residual checks, as well as comparisons between raw and standardized correlations and rolling correlation plots with regime markers. I also applied K-means clustering on return and volatility features.

Finally, I worked on extending the analysis to MS-ARCH-X models with exogenous variables. This involved setting up an estimation framework and running initial simulations to test whether LASSO can identify relevant volatility drivers. This part of the work is not finished, but it should eventually allow the regime-based analysis to be complemented.

Theoretical Background

ARCH Models

Financial returns often show volatility clustering: quiet periods followed by turbulent episodes. To capture this, Engle (1982) introduced the ARCH model. In an ARCH(q) process, today's variance depends on past squared shocks:

$$r_t = \mu + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t, \qquad \sigma_t^2 = \omega + \sum_{i=1}^{q} \alpha_i \varepsilon_{t-i}^2,$$

where z_t is white noise with unit variance. Large shocks increase future volatility, making the model well suited to financial data. In practice, high-order ARCH models are often needed.

GARCH Models

Bollerslev (1986) generalized this idea into the GARCH model, where the variance also depends on its own past values. The most common specification is the GARCH(1,1):

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2.$$

Here α measures the impact of new shocks and β measures persistence. If $\alpha + \beta$ is close to one, volatility is very persistent, which is often observed in markets. GARCH(1,1) is widely used as a baseline for volatility modeling.

Markov-Switching Extensions

Financial markets often behave differently across regimes such as Stability, Crisis, or Recovery. Markov-Switching models formalize this idea by allowing parameters to depend on an unobserved state variable S_t that follows a Markov chain. For example, in a two-regime MS-GARCH(1,1):

$$\sigma_t^2 = \omega^{(S_t)} + \alpha^{(S_t)}\varepsilon_{t-1}^2 + \beta^{(S_t)}\sigma_{t-1}^2.$$

The state S_t switches over time with given transition probabilities, allowing the volatility dynamics to differ across regimes. This framework is particularly useful to capture sudden changes in market conditions.

Exogenous Variables

Sometimes volatility is influenced by outside factors such as macroeconomic indicators or financial variables. To capture this, exogenous variables x_t can be added to the variance equation, giving ARCH-X or GARCH-X models:

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2 + \gamma x_{t-1}.$$

This extension allows additional information to help explain and predict volatility. In Markov-Switching models, exogenous variables can also affect transition probabilities between regimes.

Correlation and Standardization of Returns

Beyond volatility, it is important to study how assets move together. The correlation between two returns $r_{i,t}$ and $r_{j,t}$ is

$$\rho_{ij} = \frac{\mathrm{Cov}(r_{i,t},\, r_{j,t})}{\sqrt{\mathrm{Var}(r_{i,t})\,\mathrm{Var}(r_{j,t})}}.$$

During crises, raw correlations often rise, partly because all volatilities increase. To separate this mechanical effect from genuine co-movement, returns can be standardized by their conditional volatility:

$$z_{i,t} = \frac{r_{i,t}}{\sigma_{i,t}}.$$

Correlation matrices computed from standardized returns better reflect structural dependence across assets. Comparing raw and standardized correlations across regimes shows how much of the increase in co-movement is due to volatility itself, and how much reflects deeper changes in market linkages.

1 MS-ARCH-X Modeling and Exogenous Variables

1.1 Data Preparation and Preprocessing

We begin by modeling the volatility dynamics of the S&P 500 (SPX) index using the MS-ARCH-X framework. The function `calibrate_and_store` is used to fit the model. Our first approach is taking the SPX index as the target variable and treating all other indices as potential exogenous indicators, one at a time. The estimation relies on the following iteration thresholds:

- Hidden Markov Model (HMM): `n_iter_HMM = 1000`
- Regime-wise ARCH estimation: `niterMS = 1000`
- Exogenous variable optimization: `niterX = 100`

Prior to modeling, we preprocess the data using the function `get_integration_order(data)`, which determines the order of differencing required to make each series stationary. All indices are transformed using the natural logarithm to enforce return additivity, except for the VIX index, which is kept in its raw form due to its specific interpretability.

1.2 Model Fitting and Visualization

Once the model is fitted, outputs are stored in serialized `.pkl` files. These outputs include estimated volatility paths per regime, which are subsequently visualized and compared to the VIX index.

Each volatility trajectory is plotted individually, then overlaid with the VIX index to assess the model's realism. Refer to the Github, file $\text{spx}_v ol_v s_v ix.pdf to see the plots$.

This initial calibration phase allows us to test the performance of MS-ARCH-X under full-information conditions, leveraging all candidate exogenous variables simultaneously.

2 Regime-Based ARCH and GARCH Modeling on the S&P 500

2.1 Overview and Objectives

To motivate the introduction of regime-switching volatility models like MS-ARCH-X, we conduct a regime-based analysis of volatility dynamics in the

S&P 500 index. Specifically, we fit **ARCH(1)** and **GARCH(1,1)** models separately to periods corresponding to known economic regimes: *Stability*, *Crisis*, and *Recovery*. This allows us to observe how volatility clustering and persistence behave under different macroeconomic conditions.

2.2 Data Preparation and Regime Labeling

We use the function `load_selected_columns` to extract relevant series (SPX, VIX, and additional indicators) from Excel. Each date is then labeled with a regime tag using `add_regime_labels`, producing the dataset `SPX_with_regimes.xlsx`. The final structure includes:

- `Date` column
- `SPX Index`, `VIX Index`, and other indicators
- A new `Regime` column assigning each row to one of 15 historical macroeconomic regimes

2.3 Return Computation and Visualization

Returns are computed both in raw and log form, with a preference for log returns due to their additivity. The functions used are:

- `plot_returns_by_regime`: plots raw returns per regime
- `plot_log_returns_by_regime`: displays log return paths across regimes

The resulting visualizations reveal clear distinctions between regimes, with sharp spikes in volatility during Crisis periods and flatter behavior during Stability.

2.4 ARCH(1) Estimation per Regime

We apply an **ARCH(1)** model to each regime segment using the function `fit_arch1_per_regime`. This function computes log returns and fits the model on a regime-by-regime basis, outputting:

- μ: mean return
- α_0: baseline (unconditional) variance
- α_1: volatility clustering coefficient

Parameter analysis is performed using `analyze_arch_parameters_by_regime`, which computes means and standard deviations and visualizes the distribution via boxplots using `seaborn`.

2.5 ARCH(1) Volatility Visualization

Conditional volatility paths for each regime segment are plotted using:

- `plot_arch_volatility_paths_by_regime`

This visual analysis confirms that ARCH volatility behaves differently across regimes. However, since the ARCH model assumes independence over time, this analysis is to be interpreted regime-by-regime rather than globally.

2.6 GARCH(1,1) Estimation and Analysis

To capture more flexible volatility persistence, we extend the model to **GARCH(1,1)**. This is done using:

- `fit_garch_by_regime`, which estimates μ, ω, α_1, β_1, and computes persistence as $\alpha_1 + \beta_1$
- `plot_garch_parameter_distributions`, which produces boxplots of these parameters per regime

This deeper analysis highlights regime-dependent differences in volatility memory and shock response. Crisis periods tend to exhibit higher baseline variance and stronger clustering.

2.7 GARCH(1,1) Volatility Visualization

The conditional volatilities obtained from GARCH(1,1) are visualized using:

- `plot_garch_volatility_paths_by_regime`

Very Important point: GARCH models assume time-dependent persistence across continuous data. Fitting them per regime may violate these assumptions, so these results should be viewed illustratively.

2.8 Debugging and Robustness Checks

To investigate anomalies (such as unexpectedly low α_1 values), we conducted manual debugging:

- Compare ARCH(1) and GARCH(1,1) fits on problematic windows (e.g., 2022-01-01 to 2022-10-12)
- Use raw SPX prices to recompute returns and re-estimate parameters
- Print model parameters to assess convergence

These steps confirmed that zero or low α_1 may arise naturally in low-volatility environments or in segments with poor model fit.

2.9 Conclusion and Limitations

This regime-based volatility modeling supports the hypothesis that volatility behavior differs structurally across economic periods. However, results should be interpreted with caution due to:

- Limited sample size within each regime
- Parameter instability for short Crisis periods

To address this, we will extend this analysis to S&P 500 **sub-indices** in later sections. These offer multiple realizations per regime, enabling stronger statistical inference and more reliable validation of the MS-ARCH-X framework.

3 Regime-Aware GARCH(1,1) Modeling and Residual Diagnostics

3.1 Modeling Strategy and Motivation

In this section, we implement a **regime-aware GARCH(1,1)** model to test whether volatility parameters differ meaningfully across economic regimes. The goal is to evaluate whether this additional structure improves model fit and justifies a three-regime volatility framework.

Key questions addressed:

- Do regime-specific GARCH parameters capture structural volatility differences?
- Do standardized residuals exhibit different distributions across regimes?
- Is there evidence of volatility asymmetry within regimes?

3.2 Methodology

We build a custom GARCH(1,1) model with parameters $(\omega_k, \alpha_k, \beta_k)$ estimated separately for each regime $k \in \{\text{Stability, Crisis, Recovery}\}$. The conditional variance is computed continuously over the full return series, using the appropriate regime's parameters at each time step. This approach:

- Is not a Markov-switching GARCH, but a deterministic regime-partitioned GARCH
- Uses known regime labels to switch parameters
- Maintains a single continuous volatility series across time (otherwise would violate conditions of GARCH model)

The likelihood is optimized under constraints:

$$\omega > 0, \quad \alpha > 0, \quad \beta > 0, \quad \alpha + \beta < 1$$

Optimization is done using `scipy.optimize.minimize` with the `SLSQP` algorithm.

3.3 Residual Distribution Analysis

We analyze the standardized residuals:

$$z_t = \frac{r_t - \mu}{\sigma_t}$$

per regime using the function `analyze_regime_garch_residuals`. For each regime, we compute:

- Mean and standard deviation
- Skewness and kurtosis
- Histogram plots of the residuals

See figures in Corresponding section of the Notebook

3.4 Findings

The residual distributions show only mild differences across regimes. While Crisis segments may show slightly higher kurtosis or skew, overall the distributions remain comparable:

- here was no strong statistical or visual justification for three distinct volatility regimes based solely on residual behavior.

This means that while the regime-aware model is theoretically more flexible, the observed benefit from this additional complexity — based on standardized residuals — is limited.

3.5 Volatility Asymmetry Diagnostics

To probe deeper, we tested whether volatility differs after positive vs. negative returns — a known asymmetry in financial markets.

Method:

- Compute 40-day rolling standard deviations for positive and negative returns
- Plot them over time with regime shading
- Compute regime-wise statistics:

  - Std dev of positive returns
  - Std dev of negative returns
  - Their ratio (asymmetry indicator)

See notebook for Figures
The results are inconclusive:

- Some regimes show higher volatility after negative returns
- The asymmetry is not consistent or robust enough to serve as a regime identifier

3.6 Conclusion

While regime-aware GARCH modeling adds interpretability and theoretical richness, its empirical benefits are modest:

- Residual diagnostics do not strongly differentiate regimes
- Asymmetry effects are too noisy to serve as reliable signals

These findings suggest the need to incorporate more information — such as cross-sectional dynamics — to validate the regime-switching hypothesis more effectively.

4 Sector-Based Correlation and Regime-Aware Volatility Structure

4.1 Objective and Motivation

To deepen the justification for a three-regime volatility model, we examine how the co-movement between the S&P 500 (SPX) and its sector indices evolves across regimes. By studying both raw and standardized correlations, we aim to assess whether regime shifts impact cross-sectional dependence and whether these shifts exhibit identifiable structural patterns.

4.2 Data Sources and Preprocessing

We use two datasets (both available in the Github):

- `spx_sectors.xlsx`: contains SPX and its 10 GICS sector indices.
- `spx_stocks_clean.xlsx`: contains constituent-level data (not used in this section).

Each date is labeled according to known regime periods (`regime_periods`). This allows us to compute volatility and correlation measures on a per-regime basis.

4.3 ARCH(1) Comparison Between SPX and CAC

To compare geographic volatility behavior, we fit an `ARCH(1)` model separately to SPX and CAC returns across regimes.

Per regime, we extract:

- μ: mean return
- α_0: unconditional variance
- α_1: volatility clustering coefficient

Even though it is not theoretically rigorous, for exploratory value we compute the correlation between the conditional volatility paths of SPX and CAC as an exploratory diagnostic.

4.4 Regime-Aware GARCH Standardization

We then compute regime-specific GARCH(1,1) volatilities using:

- `fit_regime_garch`
- `compute_sigma2_regime`

For each index i, the returns are standardized as:

$$z_t^{(i)} = \frac{r_t^{(i)}}{\sqrt{\sigma_t^{2(i)}}}$$

This normalization corrects for regime-specific heteroskedasticity and enables cleaner correlation comparisons.

4.5 Correlation Matrices: Raw vs. Standardized

This is a very interesting part of my research, seeing how correlation between raw/standardised returns of different sectors. A result was a higher correlation in a crisis regime. For each regime (Stability, Crisis, Recovery), we compute:

- Raw correlation matrix from log returns
- Standardized correlation matrix using GARCH-normalized returns

These matrices are plotted using seaborn heatmaps.

[Insert Figure: Correlation matrices per regime (raw and standardized)]

4.6 Study of Correlation difference in between regimes

An interesting idea was to see the difference in correlation between the different regiems. We compute and visualize matrix differences:

$$\texttt{corr_study} = \text{raw_corr}_{\text{Crisis}} - \text{raw_corr}_{\text{Stability}}$$
$$\texttt{corr_std_study} = \text{std_corr}_{\text{Crisis}} - \text{std_corr}_{\text{Stability}}$$
$$\texttt{dev_study} = \text{raw_corr}_{\text{Crisis}} - \text{std_corr}_{\text{Crisis}}$$

These comparisons highlight:

- Elevated raw correlation in Crisis regimes
- Standardization removes part of this inflation
- Regime structure significantly affects sector co-movement

See Notebook for figures

4.7 Rolling Correlation with Regime Coloring

We implemented a regime-aware rolling correlation function in Plotly:

```
plot_rolling_correlation_plotly_regime_aware(
    df, target="SPX", index="S5MATR",
    rolling_window=40, log_returns=True, standardize=True
)
```

This produced a rolling 40-day correlation between the S&P 500 (SPX) and a selected sector index, with segments color-coded according to regime classification. Correlation lines were broken at regime transition points using the `Block ID` variable, which improves the interpretability of structural changes.

[Insert Figure: Rolling correlation plot with regime coloring]

When examining SPX versus the Materials sector (S5MATR), we observed:

- Sudden correlation drops below 0.4, often coinciding with or preceding crises
- Transitions in co-movement patterns that provide useful—though not predictive—visual cues
- Enhanced ability to interpret regime transitions through color-coded segmentation

—

4.8 Key Takeaways

This exploratory analysis of regime-aware rolling correlations highlights several important points:

- **Crisis periods** exhibit stronger and more uniform correlations across sectors
- **Volatility adjustment** via regime-aware GARCH improves correlation interpretability
- **Correlation matrices** change substantially between regimes, especially around crises
- **Rolling correlations** provide intuitive visual cues of structural breaks, even if not predictive
- Combined evidence from multiple methods **justifies the use of a 3-regime model** more convincingly than univariate volatility analysis alone

Together, these findings strongly support the *structural regime hypothesis* and motivate future extensions of the MS-ARCH-X framework toward multivariate or predictive correlation-based modeling.

5 Local GARCH Standardization

5.1 S&P 500 Sector Index Dictionary

The table below summarizes the GICS-based sector indices used in our correlation analysis:

Code	Sector Name	Description
S5CONS	Consumer Staples	Essential products — e.g., Procter & Gamble, Coca-Cola
S5ENRS	Energy	Oil, gas, and fuels — e.g., ExxonMobil, Chevron
S5HLTH	Health Care	Pharma, biotech, devices — e.g., Johnson & Johnson, Pfizer
S5FINL	Financials	Banks, insurance — e.g., JPMorgan, Goldman Sachs
S5UTIL	Utilities	Electric, gas, water — e.g., Duke Energy, NextEra
S5MATR	Materials	Chemicals, construction — e.g., DuPont, Int'l Paper
S5INFT	Info Technology	Software, hardware — e.g., Apple, Microsoft
S5INDU	Industrials	Transport, aerospace — e.g., Boeing, FedEx
S5TELS	Communication	Media, telecom — e.g., Google, Meta, Verizon
S5COND	Cons. Discretionary	Retail, luxury — e.g., Amazon, Nike, Starbucks

Table 1: Sector Index Codes and Descriptions

5.2 Local GARCH(1,1) Standardization by Regime

To further validate the regime-dependent structure in sector correlations, we refined our approach by fitting GARCH(1,1) models **within each regime window** separately for each index. This eliminates look-ahead bias and better captures economic segmentation.

Procedure:

- For each sector and regime (e.g., Crisis 2007–2009):
  - Compute log returns
  - Fit a GARCH(1,1) model locally
  - Reconstruct conditional volatilities σ_t
  - Standardize returns using $z_t = \frac{r_t}{\sigma_t}$

5.3 Refined Correlation Matrix Estimation

Using the locally standardized returns, we recomputed:

- Raw correlation matrices by regime
- GARCH-standardized correlation matrices by regime

This method improves robustness by ensuring that standardization only uses regime-local information. Results support the hypothesis that:

- Co-movement increases during Crisis periods

- Standardization corrects volatility-driven inflation of correlations
- Regimes imprint distinguishable structures on cross-sectional correlation patterns

6 Empirical Regime Discovery: Volatility–Return Clustering

In thi part of our analysis, we investigate whether regime labels can be *discovered algorithmically*—without imposing predefined time periods—by applying clustering techniques in return–volatility space.

6.1 K-Means Clustering on Return–Volatility Features

Using the S&P 500 (SPX) series, we computed two rolling features:

- 50-day average return
- Annualized 50-day rolling volatility

After standardizing these features with `StandardScaler`, we applied K-Means clustering with $k = 3$. The expectation was that the algorithm would identify groups corresponding to:

- **Stability:** low volatility, neutral returns
- **Crisis:** high volatility, often with negative returns
- **Recovery:** high volatility, but positive returns

```
kmeans = KMeans(n_clusters=3)
labels = kmeans.fit_predict(X_scaled)
```

The resulting clusters, visualized in return–volatility space, revealed clearly separable groups. Although no economic meaning was assigned a priori, the location and shape of clusters closely resembled the expected regime archetypes.

6.2 Comparison with Manually Assigned Regimes

To assess alignment with our predefined regime chronology, we repeated the analysis using manually assigned regime labels:

- The dataset was reloaded and annotated with regime periods.
- For clarity, only *Crisis* and *Recovery* regimes were retained, omitting *Stability* which dominates the sample.

- Observations were plotted in return–volatility space, color-coded by regime.

abel=regime)

The comparison revealed that:

- Manually assigned Crisis and Recovery periods mapped coherently into distinct regions of return–volatility space.
- These regions corresponded closely to two of the three K-Means-identified clusters.
- Stability periods, being frequent and diffuse, were distributed across multiple clusters and were thus less identifiable from data alone.

6.3 Interpretation

This clustering exercise provides a valuable sanity check on the regime specification. Despite the absence of economic or temporal information, the data-driven clustering method naturally reproduced two of the three hypothesized regimes. This confirms the Crisis and Recovery structure present in SPX return–volatility dynamics. Stability, by contrast, lacks a sharp signature in two-dimensional feature space, consistent with its more diffuse and persistent nature.

Takeaway: While K-Means clustering cannot independently recover all regimes—particularly Stability—it validates the existence of regime structure in financial time series. This strengthens support for the three-regime hypothesis (Stability, Crisis, Recovery) and illustrates the value of combining statistical learning with economic intuition.

7 GARCH(1,1) Parameter Behavior Across Regimes and Indices

We now investigate whether the parameters of the GARCH(1,1) model vary systematically across economic regimes—Crisis, Stability, and Recovery—by extending the analysis beyond the aggregate SPX index.

7.1 From SPX to Sector Indices

To mitigate the limitations of sample size when using SPX alone, we incorporated a broader dataset, `spx_sectors.xlsx`, which contains daily price series for the SPX and its eleven sector sub-indices. Each index was segmented using the same regime chronology, ensuring comparability of results across sectors.

7.2 Fitting GARCH(1,1) Models by Regime

For each index, daily log returns were computed and a GARCH(1,1) specification was estimated separately within each regime window. The fitted models yielded the following parameters:

- μ: unconditional mean return
- ω: long-run variance
- α_1: sensitivity to short-term shocks
- β_1: persistence of volatility
- $\alpha_1 + \beta_1$: overall volatility persistence, summarizing long-term memory

Regimes with fewer than 100 return observations were excluded to avoid unreliable estimates.

7.3 Visualizing Parameter Distributions

To summarize results across indices, we pooled all fitted models and plotted parameter distributions using Seaborn swarmplots, grouped by regime. This visualization highlighted both the dispersion and clustering of parameter estimates across economic conditions. In addition, we reported the number of models contributing to each regime group, providing transparency regarding the robustness of the distributional patterns.

Takeaway: The cross-sectional distribution of GARCH(1,1) parameters across sectors and regimes provides insight into how volatility dynamics adapt to changing economic environments, thereby offering further evidence for regime-specific modeling.

8 Ongoing Study: LASSO in ARCH-X and Transition to MS-ARCH-X

8.1 ARCH-X Simulation Study with LASSO Selection

8.1.1 Objective

Assess whether the LASSO method can identify the correct exogenous variables in a simulated ARCH(1)-X volatility model.

8.1.2 Methodology

We simulate 10 AR(1) exogenous variables, of which only two influence volatility. The conditional variance is given by:

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \sum_{j=1}^{10} \gamma_j X_{j,t}, \quad \gamma_j = 0 \text{ except for two active variables.}$$

Returns are generated as:

$$r_t \sim \mathcal{N}(0, \sigma_t^2).$$

We then apply `lasso_path` from `sklearn` to scan across values of λ and select the most relevant variables. Repeated simulations yield a success rate defined as the frequency with which LASSO correctly identifies the two active variables.

8.2 Limitations of LASSO for MS-ARCH-X

While effective in ARCH-X, LASSO faces theoretical limitations in the MS-ARCH-X framework:

- `lasso_path` assumes a linear regression setting, whereas MS-ARCH-X is nonlinear with hidden states.
- Variable selection based on Viterbi-estimated regimes (from a model without exogenous inputs) is biased:
  - The state path S_t is conditional on a reduced model ignoring exogenous influences.
  - This leads to misclassification of regimes when exogenous variables affect volatility or transitions.
  - A regime-wise LASSO applied to these misclassified states may discard relevant variables.
- Any variable selection performed on a misspecified model undermines theoretical validity.

8.3 Impact of Sample Size on LASSO Performance

8.3.1 Objective

Evaluate how sample size T affects LASSO's success in recovering the true exogenous variables in an ARCH(1)-X setting.

8.3.2 Procedure

- Simulate $k = 10$ AR(1) exogenous variables, with only two active in volatility dynamics.
- Active coefficients fixed at $[0.8, -0.5]$.
- Vary T from 500 to 10,000 in steps of 100.
- For each T, run 1000 independent trials with `single_lasso_trial`.
- Compute the success rate: proportion of trials where exactly the two active variables are selected.

See Notebook for Plot, but get close to 99 percent with 3000 points

8.4 Initialization Strategies for MS-ARCH-X

We consider two valid approaches to initializing the MS-ARCH-X model:

Option 1: Initialization via `get_initial_conditions` (Recommended)

- Call `get_initial_conditions(eps, initial_params, d=2, q=1)`.
- Provide exogenous variables through:
  - `'exogen_additive'` for volatility dynamics.
  - `'exogen_transition'` for regime transitions.
- Generates a simplified model to obtain consistent starting parameters.

Option 2: Zero-Padded Initialization from MS-ARCH

- Estimate an MS-ARCH without exogenous variables.
- Extend the estimated parameter vector with zeros for exogenous coefficients.
- Use this as the starting point for MS-ARCH-X estimation.

8.5 Next Steps

The immediate goal is to implement `get_initial_conditions` on simulated or real data, including the fields `'exogen_additive` and/or `'exogen_transition'`. No hard variable selection will be imposed at this stage. Initial experiments will use a limited number of AR(1) exogenous variables, testing additive and transition effects separately. This will lay the foundation for stable estimation of the complete MS-ARCH-X model.